

Abhigyan Kumar

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PROFESSIONAL SUMMARY

Backend and Data Science enthusiast with hands-on experience building REST APIs, database-driven applications, and machine learning solutions using Python. Experienced in FastAPI, Flask, SQL databases, data analysis, machine learning, and backend system design. Built and deployed end-to-end applications involving authentication, concurrency, data processing, predictive modeling, and interactive analytics. Strong foundation in Data Structures and Algorithms, Object-Oriented Programming, and software development, with a growing focus on integrating AI into software applications.

EDUCATION

The National Institute of Engineering (NIE), Mysore
B.E. in Information Science and Engineering

2023 – 2027
CGPA: 8.2

TECHNICAL SKILLS

Languages: Python, C, C++, SQL

Core CS: Data Structures, Algorithms, OOP, Operating System, DBMS, Computer Networks

Libraries / Frameworks: Pandas, NumPy, scikit-learn, Django, FastAPI, Flask

Tools & Services: Linux, Git, GitHub, Docker

PROJECTS

AaharSetu : Food Waste Redistribution Platform | *FastAPI, PostgreSQL, WebSockets, Leaflet* | GitHub

- Engineered high-concurrency food redistribution backend using FastAPI and SQLAlchemy 2.0 (asyncpg), eliminating race conditions and double-claims via PostgreSQL pessimistic row-level locking (SELECT ... FOR UPDATE).
- Implemented geospatial search & routing using the Haversine great-circle distance formula, enabling real-time proximity-based radius filtering across active surplus food listings.
- Secured platform access with Role-Based Access Control (RBAC) across 3 user tiers (donor, receiver, admin), leveraging Bcrypt password hashing and dual-mode JWT authentication (Bearer tokens & httpOnly secure cookies).
- Built an end-to-end automated test suite with 52+ async test cases using Pytest, pytest-asyncio, and httpx, validating transaction isolation, authorization boundaries, and edge cases.

Fake News Detector | *Machine Learning, Flask, NLP* | GitHub | Live

- Built a full-stack ML web app that classifies news articles as Fake or Real using NLP techniques.
- Implemented TF-IDF vectorization and Logistic Regression achieving binary classification with real-time predictions.
- Deployed on Render with a Flask backend, containerized via Docker, and served a clean HTML/CSS frontend.

CERTIFICATIONS

Data Structures in C – Coursera

Linux for Developers – Coursera

Data Visualization with Python – Coursera

ADDITIONAL

Soft Skills: Team Collaboration, Time Management, Fast Learner

Interests: Generative AI, Competitive Programming, Watching Sports & Anime, Travelling